



The observable universe

Check out the amazing infographics of Pablo Carlos Budassi in high resolution, including the universe map. <https://dgit.in/jan20-11>



Planck results

The combined results of the Planck observations, which changed how we see the universe. <https://dgit.in/jan20-12>

light-years. The universe is also estimated to be around 13.799 billion years old. There are stars and galaxies that we will never be able to see. The farthest stars and galaxies that we can currently observe, will fade away in 6 billion years, after freezing in their locations in the sky, and becoming progressively redder and darker. As the

they started, a spaceship moving in an absolutely straight line would eventually end up where they started. However, the surface of the sphere is not the only possible shape of such a universe that folds in on itself. Another possible shape is the surface of a torus, or a donut shape. Think of these as three dimensional analogues of the mobius strip. One of the most interesting of these possible shapes is that the universe is shaped like a Klein bottle, an object that has a single surface without an inside or an outside, but one that can still hold liquids within it. Now if the universe were shaped like the surface of the sphere, it would be a simply connected universe. If there is a source of light and an observer, there is just one path that the light can take to reach the observer. A Klein bottle or torus shaped universe however, are what is known as "multiply connected". This means that the light from a source, can reach the observer through multiple paths. This would be similar to standing in a hall of mirrors, with replications of the same object. We might be seeing two stars in the sky in such a universe, when actually both the stars are one and the same. Whether the universe is simply connected or multiply connected is deeply linked to another aspect of the geometry, known as curvature.

Now the curvature of the universe is different from the curvature that we experience in everyday lives. A positive curvature is spherical, a negative curvature is shaped like a saddle, while a flat surface is well, like a piece of paper.

Now bear in mind that none of these geometries have edges, and all of them are finite. A simple way to understand these different curvatures is to imagine a set of parallel lines. If the parallel lines diverge because of the structure of space itself, then the universe has a negative curvature. If the lines meet themselves, or tend to converge, then the universe has a positive curvature. If however, the lines remain parallel forever, the universe is considered to be flat. You can find out the curvature of a shape by drawing parallel lines on it. For example, a torus has a flat curva-



A Klein bottle

universe expands, the amount of it that we can see actually reduces with the passage of time. The entire universe, even the parts beyond the observable universe, is known as the global universe. It is difficult to figure out the shape of such a universe, and it is as such an unanswered question of cosmology. However, physicists can think of the range of possible options, and pick the ones that best match the observations by astronomical instruments.

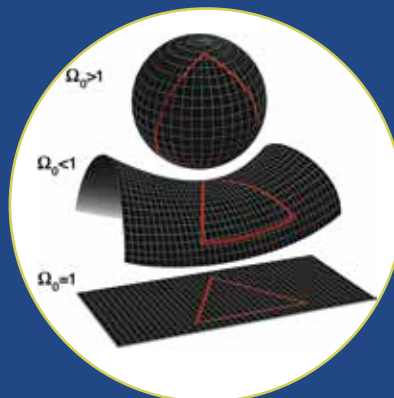
Scientists are now fairly certain that the universe is finite, because it is not entirely on fire. Now the question is what happens if you travel to the edge of the universe. The question itself is somewhat illogical. The universe encompasses everything that exists. Anything outside the universe is paradoxically, also a part of the universe. Although in everyday experience, most of the objects we encounter have an edge, it is difficult for physicists to even imagine a universe with an edge. There are some things that do not have an edge, the simplest example being the surface of the Earth. Just as someone walking endlessly in a single direction over land or water would end up where



An illustration of the observable universe in logarithmic scale

ture, just like a sheet of paper, because eventually, the two parallel lines drawn on it in any direction will meet each other perfectly, without diverging or converging. Measurements from the Planck spacecraft based on the density of matter in the observable universe, indicates that the Universe is flat. Now while intuitively it may seem like the only way that a flat universe is also simply connected would be for it to be infinite, but this is not the case.

While we do not know the shape of the universe, we do have evidence to suggest that it is flat and without an edge. We do not know conclusively if it is simply connected or multiply connected. Now remember that spaceship that heads out in a straight line and ends up where it started? How long the trip would take, would decide the size of the universe. That is another thing, we simply don't have the answer to. 



From top to bottom, universes with positive, negative and flat curvatures